

- BC 21. The value of the derivative of $y = \frac{\sqrt[3]{x^2+8}}{\sqrt[4]{2x+1}}$ at $x = 0$ is

1993

SC

- (A) -1 (B) $-\frac{1}{2}$ (C) 0 (D) $\frac{1}{2}$ (E) 1

- BC 6. The line normal to the curve $y = \sqrt{16-x}$ at the point $(0, 4)$ has slope

1997

NC

- (A) 8 (B) 4 (C) $\frac{1}{8}$ (D) $-\frac{1}{8}$ (E) -8

- BC 5. If f and g are twice differentiable and if $h(x) = f(g(x))$, then $h''(x) =$

1998

NC

- (A) $f''(g(x))[g'(x)]^2 + f'(g(x))g''(x)$
(B) $f''(g(x))g'(x) + f'(g(x))g''(x)$
(C) $f''(g(x))[g'(x)]^2$
(D) $f''(g(x))g''(x)$
(E) $f''(g(x))$

- BC 8. If $h(x) = f^2(x) - g^2(x)$, $f'(x) = -g(x)$, and $g'(x) = f(x)$, then $h'(x) =$

1969

NC

- (A) 0 (B) 1 (C) $-4f(x)g(x)$
(D) $(-g(x))^2 - (f(x))^2$ (E) $-2(-g(x) + f(x))$

- BC 39. Let f and g be differentiable functions such that

1973

NC

$$\begin{array}{lll} f(1) = 2, & f'(1) = 3, & f'(2) = -4, \\ g(1) = 2, & g'(1) = -3, & g'(2) = 5. \end{array}$$

If $h(x) = f(g(x))$, then $h'(1) =$

- (A) -9 (B) -4 (C) 0 (D) 12 (E) 15